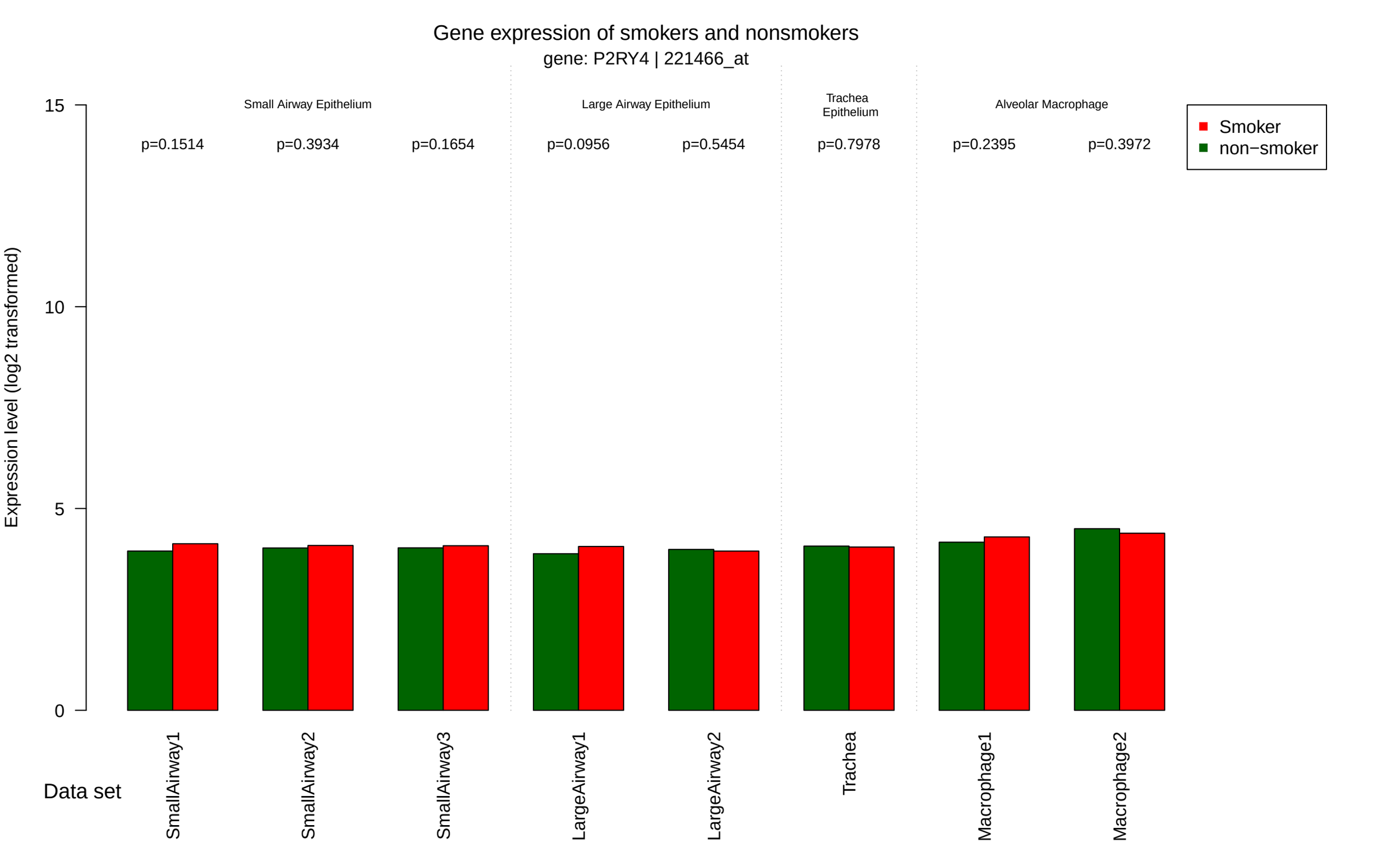
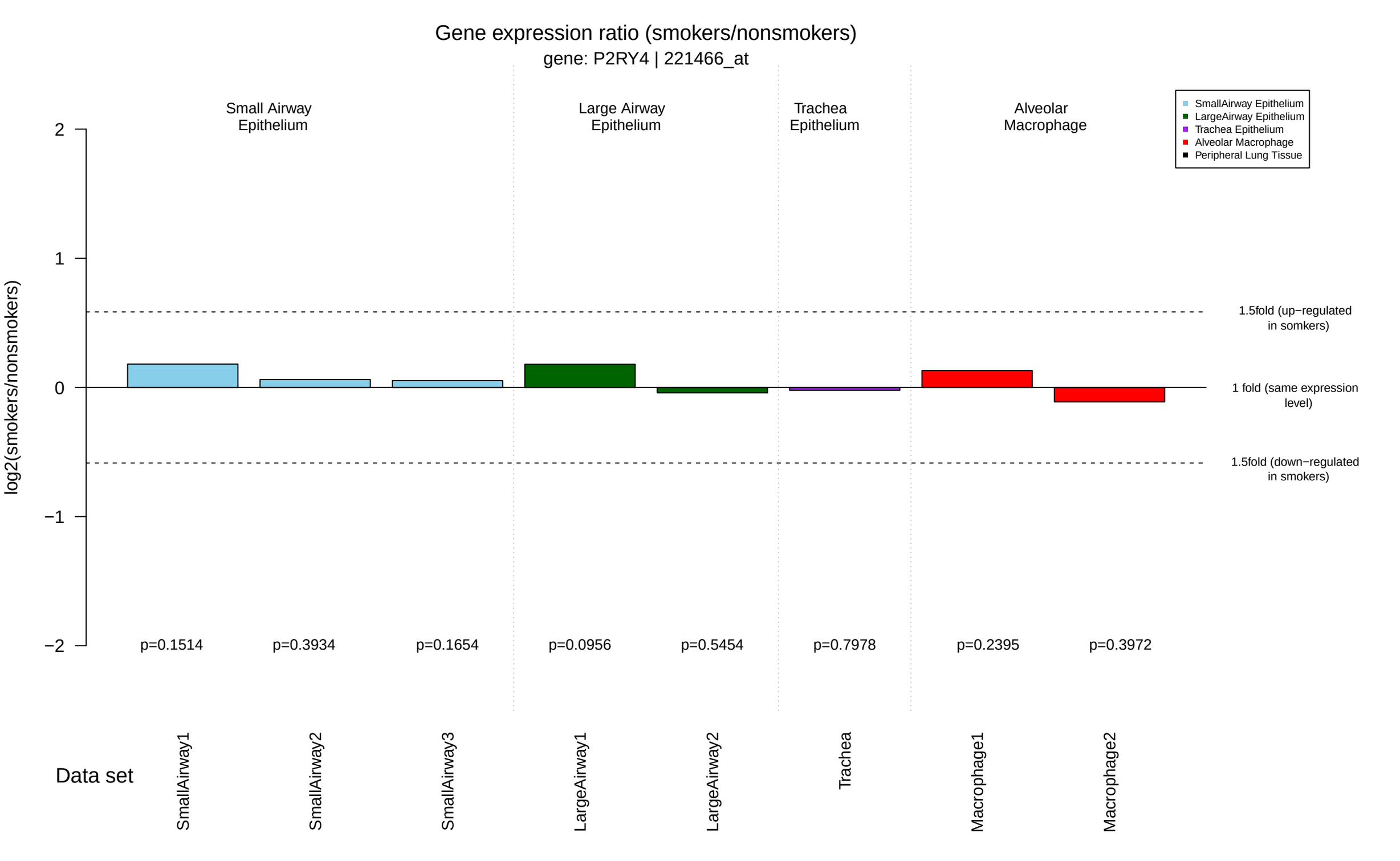
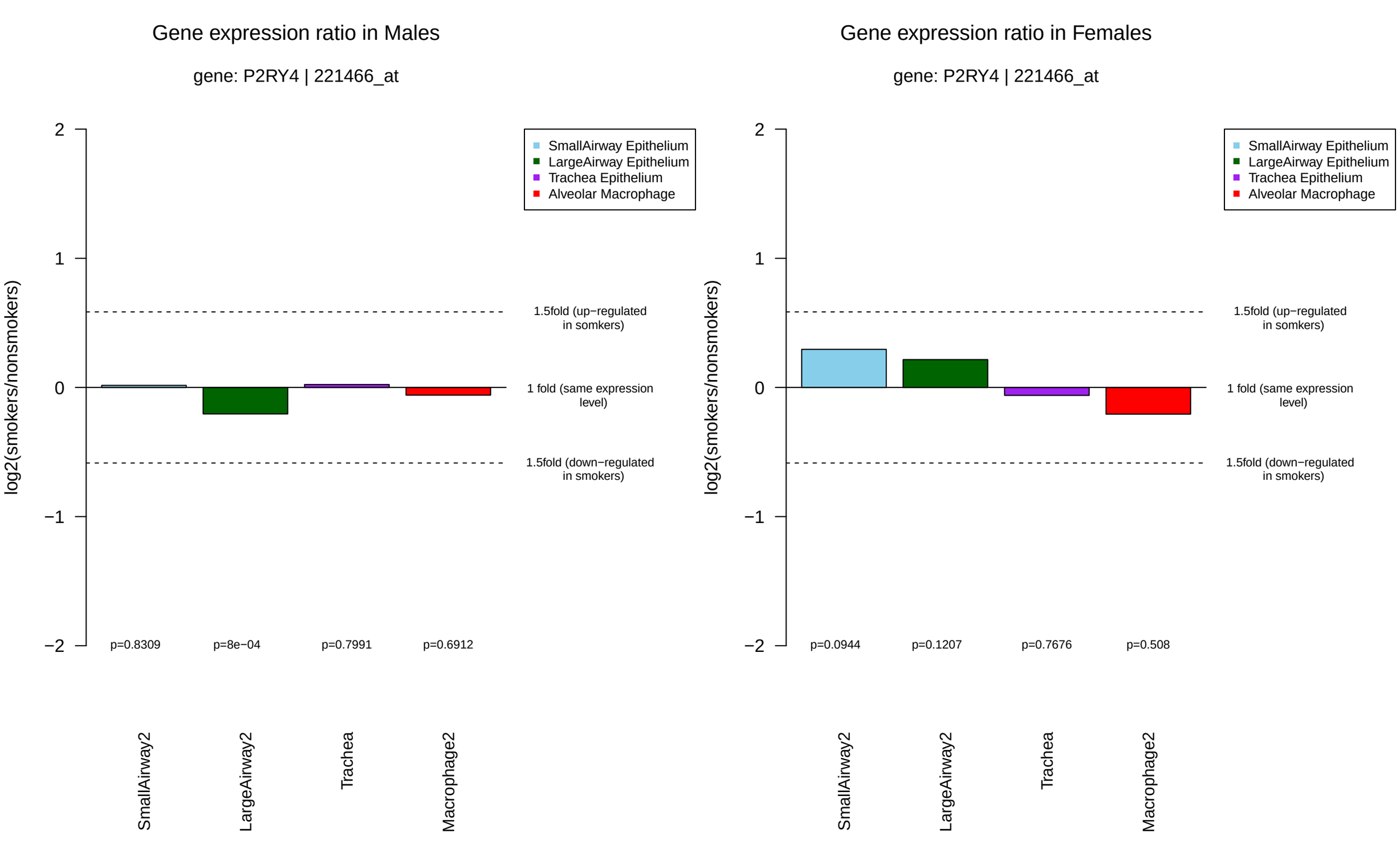
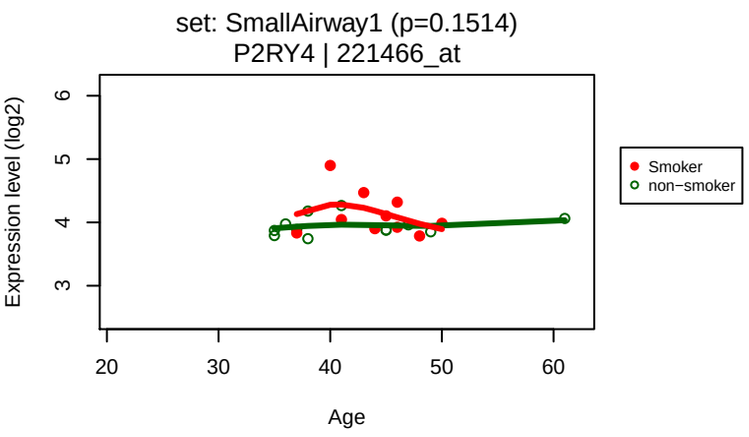
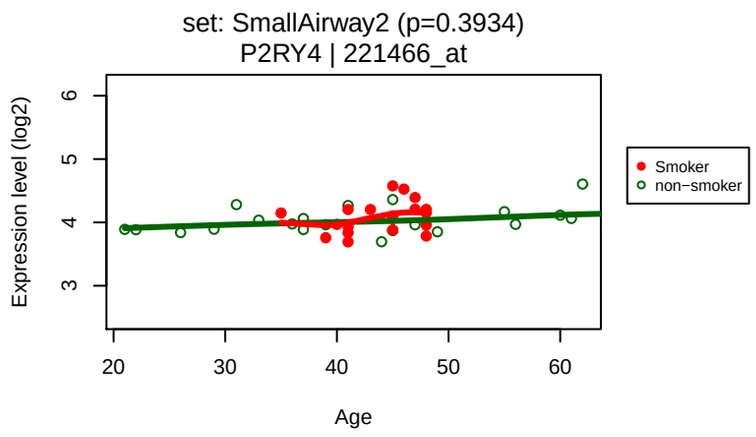
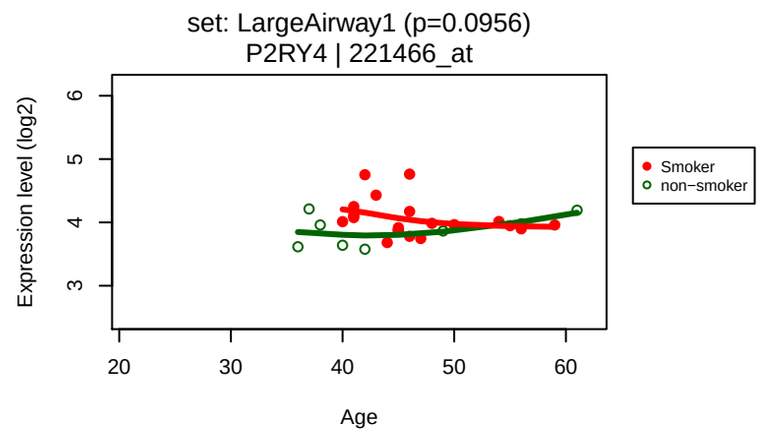

Gene expression of smokers and nonsmokers
gene: P2RY4 | 221466_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1514
p=0.3934
p=0.1654
p=0.0956
p=0.5454
p=0.7978
p=0.2395
p=0.3972
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

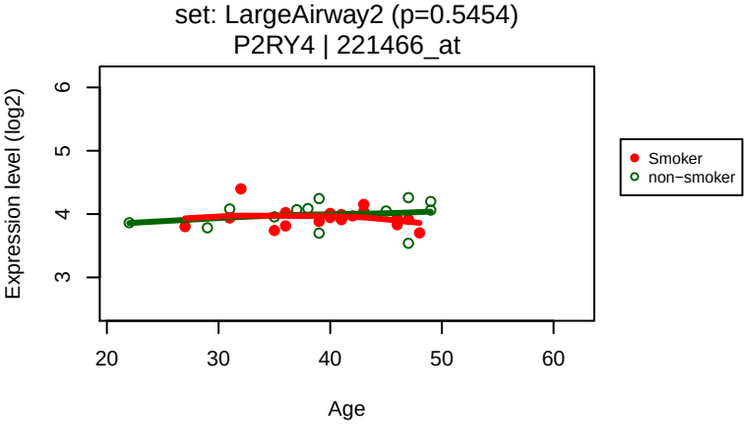

Gene expression ratio (smokers/nonsmokers)
gene: P2RY4 | 221466_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1514
p=0.3934
p=0.1654
p=0.0956
p=0.5454
p=0.7978
p=0.2395
p=0.3972
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

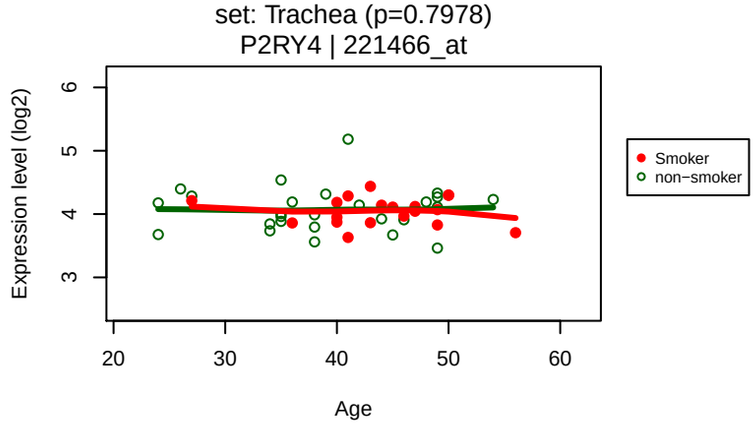

Gene expression ratio in Males
gene: P2RY4 | 221466_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8309
p=8e−04
p=0.7991
p=0.6912
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P2RY4 | 221466_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0944
p=0.1207
p=0.7676
p=0.508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

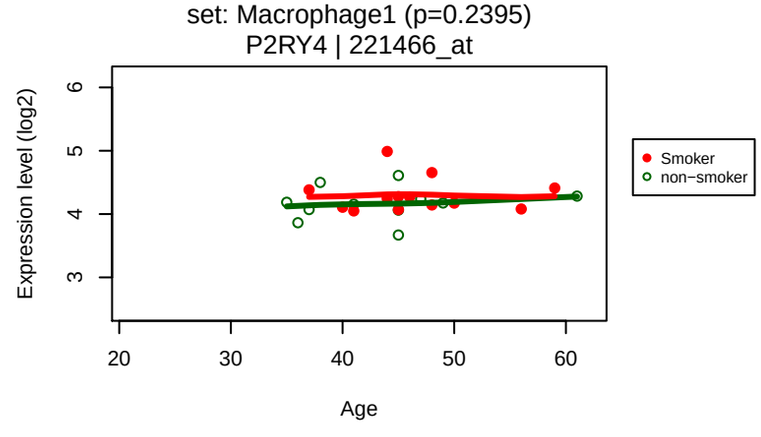

set: SmallAirway1 (p=0.1514)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker

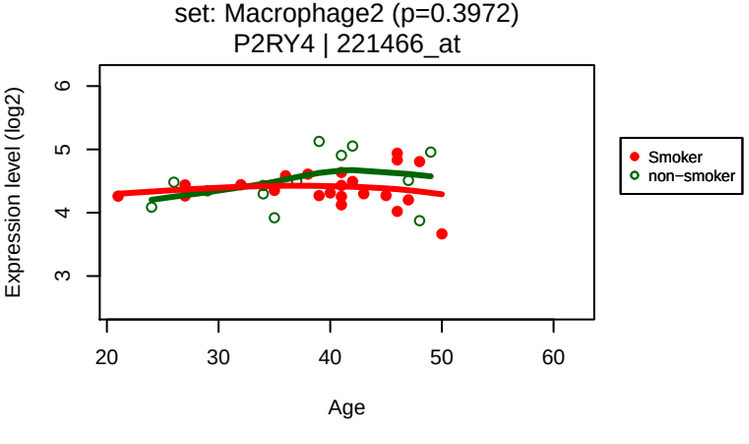

set: SmallAirway2 (p=0.3934)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker

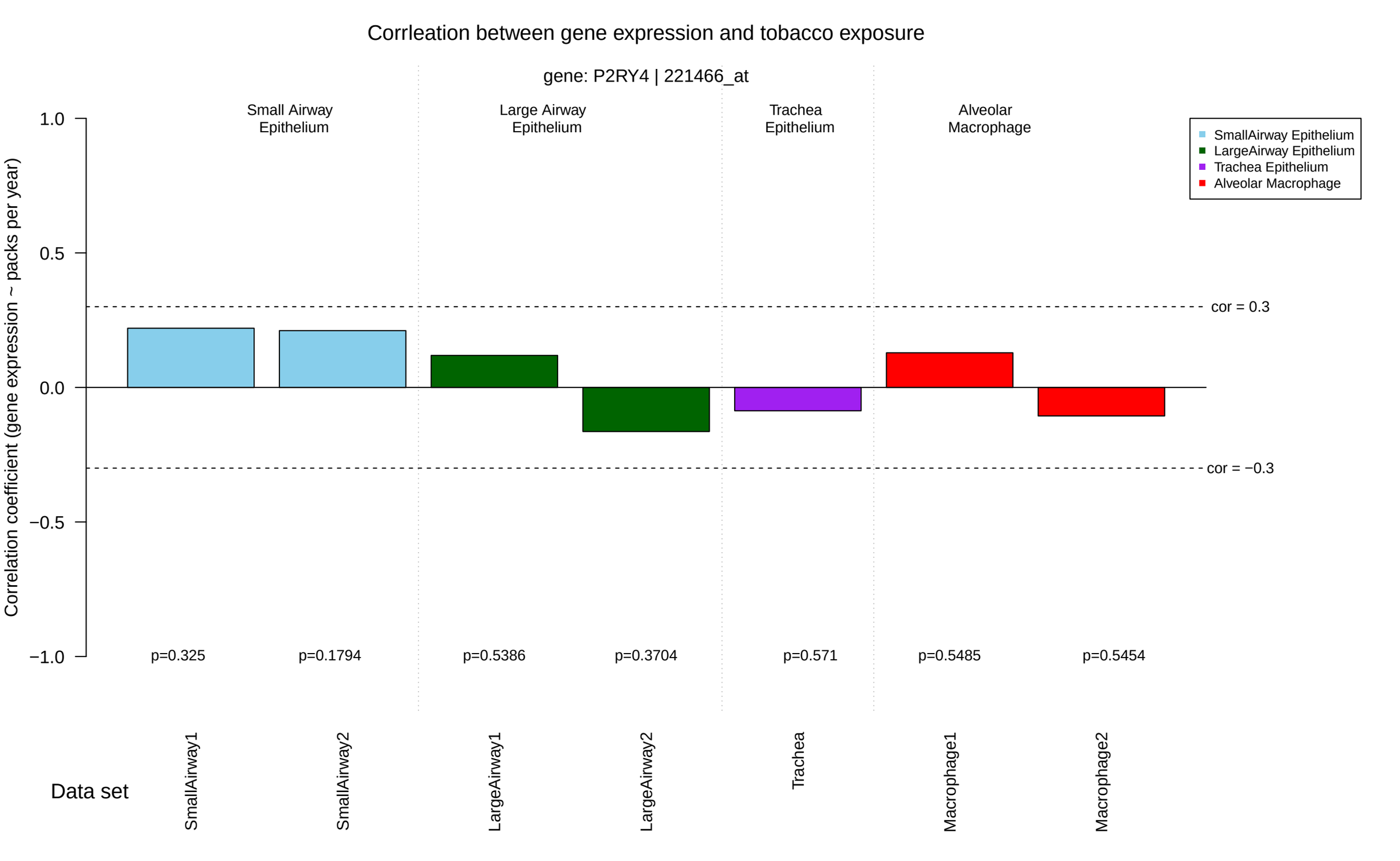

set: LargeAirway1 (p=0.0956)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5454)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7978)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2395)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3972)
P2RY4 | 221466_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P2RY4 | 221466_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.325
p=0.1794
p=0.5386
p=0.3704
p=0.571
p=0.5485
p=0.5454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2